Analysis of Partial-Share (PS) Algorithm

Derivation Note

Algorithm 1 Barrier Partial-Share Algorithm

Input: Tree $\mathcal{T}$, shared leaf set $\mathcal{L}_{\text{shr}}$, non-leaf node gate $g(i) \in \{\text{share}, \text{unshare}\}$, learning rate η, exploration base rate ϵ.
Notation: For a node i, let $\mathcal{L}(i)$ denote the set of leaf descendants of i.
Pre-processing: Label a non-leaf node i as *Safe* if

$$\mathcal{L}(i) \subseteq \mathcal{L}_{\text{shr}} \quad \text{and} \quad g(u) = \text{share}, \ \forall \text{ non-leaf } u \in \text{subtree}(i);$$

otherwise label it as *Risky*.
Initialize:
  $\theta_\ell \leftarrow 0, \ \forall \ell \in \mathcal{L}_{\text{shr}}$
  $\theta[i,j] \leftarrow 0, \ \forall$ risky node i and $j \in Children(i)$
  For every safe node u:

$$W[u] \leftarrow \sum_{\ell \in \mathcal{L}(u)} \exp(\eta\theta_\ell)$$

for each round $t = 1, \ldots, T$ **do**
  $i \leftarrow root$, $\Pi_t(root) \leftarrow 1$
  while i is not a leaf **do**
    if i is Safe **then**
      for each $j \in Children(i)$ **do**

$$p_t(j \mid i) \leftarrow \frac{W[j]}{W[i]}$$

      end for
    else

$$\epsilon_i \leftarrow \epsilon \cdot \mathbf{1}\{Children(i) \not\subseteq \mathcal{L}\}$$

      Randomly select $m[i] \in \{U, E\}$ with $\Pr(m[i] = U) = \epsilon_i$
      for each $j \in Children(i)$ **do**
        if $m[i] = U$ **then**

$$p_t(j \mid i) \leftarrow \frac{1}{|Children(i)|}$$

        else

$$p_t(j \mid i) \leftarrow \frac{\exp(\eta\theta[i,j])}{\sum_{k \in Children(i)} \exp(\eta\theta[i,k])}$$

        end if
      end for
    end if
    Sample $j_t \sim p_t(\cdot \mid i)$ and record the selected edge (i, j_t)
    $\Pi_t(j_t) \leftarrow \Pi_t(i)\, p_t(j_t \mid i)$
    $i \leftarrow j_t$
  end while
  Let sampled leaf be $\ell_t \leftarrow i$. Observe terminal cost c_t.
  Update risky ancestors on the sampled path:
  For each risky node u on the sampled path whose selected child is v:

$$\theta[u,v] \leftarrow \theta[u,v] - \frac{c_t}{\Pi_t(u)\, p_t(v \mid u)}$$

  if $\ell_t \in \mathcal{L}_{\text{shr}}$ **then**
    $val_{\text{old}} \leftarrow \exp(\eta\theta_{\ell_t})$
    $\theta_{\ell_t} \leftarrow \theta_{\ell_t} - \dfrac{c_t}{\Pi_t(\ell_t)}$
    $\Delta_t \leftarrow \exp(\eta\theta_{\ell_t}) - val_{\text{old}}$
    Back-propagation: Propagate Δ_t upward along ancestors until the first node u with $g(u) =$ unshare, and stop there.
  end if
end for

Algorithm 2 Risky-PS Algorithm

Input: Tree $\mathcal{T}$, shared leaf set $\mathcal{L}_{\text{shr}}$, learning rate η, exploration base rate ϵ.
Notation: For a node i, let $\mathcal{L}(i)$ denote the set of leaf descendants of i.
Pre-processing: Label a non-leaf node i as *Safe* if $\mathcal{L}(i) \subseteq \mathcal{L}_{\text{shr}}$, and *Risky* otherwise.
Initialize:
  $\theta_\ell \leftarrow 0,\ \forall \ell \in \mathcal{L}_{\text{shr}}$
  $\theta[i,j] \leftarrow 0,\ \forall$ risky node i and $j \in Children(i)$
  For every safe node u:

$$W[u] \leftarrow \sum_{\ell \in \mathcal{L}(u)} \exp(\eta\theta_\ell)$$

for each round $t = 1, \ldots, T$ **do**
  $i \leftarrow root$, $\Pi_t(root) \leftarrow 1$
  while i is not a leaf **do**
    if i is Safe **then**
      for each $j \in Children(i)$ **do**

$$p_t(j \mid i) \leftarrow \frac{W[j]}{W[i]}$$

      end for
    else

$$\epsilon_i \leftarrow \epsilon \cdot \mathbf{1}\{Children(i) \not\subseteq \mathcal{L}\}$$

      Randomly select $m[i] \in \{U, E\}$ with $\Pr(m[i] = U) = \epsilon_i$
      for each $j \in Children(i)$ **do**
        if $m[i] = U$ **then**

$$p_t(j \mid i) \leftarrow \frac{1}{|Children(i)|}$$

        else

$$p_t(j \mid i) \leftarrow \frac{\exp(\eta\theta[i,j])}{\sum_{k \in Children(i)} \exp(\eta\theta[i,k])}$$

        end if
      end for
    end if
    Sample $j_t \sim p_t(\cdot \mid i)$ and record the selected edge (i, j_t)
    $\Pi_t(j_t) \leftarrow \Pi_t(i)\, p_t(j_t \mid i)$
    $i \leftarrow j_t$
  end while
  Let sampled leaf be $\ell_t \leftarrow i$. Observe terminal cost c_t.
  Update risky ancestors on the sampled path:
  For each risky node u on the sampled path whose selected child is v:

$$\theta[u,v] \leftarrow \theta[u,v] - \frac{c_t}{\Pi_t(u)\, p_t(v \mid u)}$$

  if $\ell_t \in \mathcal{L}_{\text{shr}}$ **then**
    $val_{\text{old}} \leftarrow \exp(\eta\theta_{\ell_t})$
    $\theta_{\ell_t} \leftarrow \theta_{\ell_t} - \dfrac{c_t}{\Pi_t(\ell_t)}$
    $\Delta_t \leftarrow \exp(\eta\theta_{\ell_t}) - val_{\text{old}}$
    Back-propagation: Propagate Δ_t to W along safe ancestors.
  end if
end for

Algorithm 3 Full-Share Algorithm (Recursive Aggregation)

Input: Tree $\mathcal{T}$, learning rate η
Initialize:
  $\theta_\ell \leftarrow 0, \forall \ell \in \mathcal{L}$.
  $W[u] \leftarrow \sum_{\ell \in \text{subtree}(u)} \exp(\eta \cdot \theta_\ell), \forall u \in \mathcal{T}$.
for each round $t = 1, \ldots, T$ **do**
  Selection:
  $i \leftarrow root$
  while i is not a leaf **do**
    Select child j from $Children(i)$ with probability:

$$P(j|i) = \frac{\sum_{\ell \in \text{subtree}(j)} \exp(\eta \cdot \theta_\ell)}{\sum_{\ell \in \text{subtree}(i)} \exp(\eta \cdot \theta_\ell)} = \frac{W[j]}{W[i]}$$

    $i \leftarrow j$
  end while
  Let leaf $\ell_t \leftarrow i$.
  Observe cost c_t at leaf ℓ_t.
  Update:
  $\theta_{old} \leftarrow \theta_{\ell_t}$
  $\theta_{\ell_t} \leftarrow \theta_{\ell_t} - \frac{c_t}{\Pi_t(\ell_t)}$
  Propagate Change:
  Calculate weight change at leaf:

$$\Delta = \exp(\eta \cdot \theta_{\ell_t}) - \exp(\eta \cdot \theta_{old})$$

  $curr \leftarrow \ell_t$
  while $curr \neq root$ **do**
    $W[curr] \leftarrow W[curr] + \Delta$
    $curr \leftarrow Parent(curr)$
  end while
  $W[root] \leftarrow W[root] + \Delta$
end for

1 Barrier Recursive Root Regret Bound for Risky-PS

Definitions

For any node i, define the cumulative expected loss of the algorithm starting from i by

$$Y_\eta[i,T] \triangleq \sum_{t=1}^{T} \mathbb{E}[\, y_\eta[i,t]\,].$$

Let $Y_*[i,T]$ denote the cumulative loss of the optimal stationary policy starting from node i, defined recursively as

$$Y_*[i,T] = \begin{cases} \sum_{t=1}^{T} c_t(i), & \text{if } i \in \mathcal{L}, \\ \min_{j\in C_i} Y_*[j,T], & \text{otherwise.} \end{cases}$$

For any non-leaf node i, define its local branching factor and optimal-child set by

$$d_i \triangleq |C_i|, \qquad C_i^* \triangleq \arg\min_{j\in C_i} Y_*[j,T].$$

For each non-leaf node i, let

$$g(i) \in \{\text{share}, \text{unshare}\}$$

denote its telemetry gate.

We say that a non-leaf node i is *safe* if

$$\mathcal{L}(i) \subseteq \mathcal{L}_{\text{shr}} \qquad \text{and} \qquad g(u) = \text{share},\ \forall \text{ non-leaf } u \in \text{subtree}(i).$$

Otherwise, i is called *risky*. To make the risky depth precise, define the *risky height* of each node i recursively by

$$R(i) \triangleq \begin{cases} 0, & \text{if } i \text{ is safe,} \\ 1, & \text{if } i \text{ is risky and } C_i \subseteq \mathcal{L}, \\ 1 + \max_{j\in C_i} R(j), & \text{if } i \text{ is risky and } C_i \not\subseteq \mathcal{L}. \end{cases}$$

We then define the *risky skeleton depth* of the whole tree as

$$R \triangleq R(r),$$

where r is the root. R is the maximum number of risky nodes that can appear on any root-to-leaf path before the path enters a barrier-free fully shared suffix.

For risky nodes, we use the R-adjusted parameters

$$\eta_i^{\text{risk}} = T^{-\frac{R}{R+1}}, \qquad \epsilon_i = \begin{cases} 0, & C_i \subseteq \mathcal{L}, \\ d_i\, T^{-\frac{1}{R+1}}, & C_i \not\subseteq \mathcal{L}. \end{cases}$$

For a safe node i, let

$$N_i \triangleq |\mathcal{L}(i)|$$

be the number of leaves in the safe subtree rooted at i. We upper bound its local contribution by the standard N_i-armed Exp3 rate over at most T rounds, using

$$\eta_i^{\text{shr}} = \sqrt{\frac{\ln N_i}{T\, N_i}}.$$

Lemma RR-1 (Safe-Subtree Exp3 Equivalence). For any safe node i, let

$$N_i \triangleq |\mathcal{L}(i)|$$

denote the number of leaves in the safe subtree rooted at i. Then the recursive selection rule of the full-share algorithm restricted to the subtree of i is equivalent to running an N_i-armed Exp3 forecaster directly on the leaf set $\mathcal{L}(i)$.

More precisely, if the leaf weights are

$$w_t(\ell) \triangleq \exp\big(\eta_i^{\mathrm{shr}}\theta_{\ell,t}\big), \qquad \ell \in \mathcal{L}(i),$$

and the recursive aggregate at any node u is

$$W_t[u] \triangleq \sum_{\ell \in \mathrm{subtree}(u)} w_t(\ell),$$

then for any leaf $\ell \in \mathcal{L}(i)$, the probability that the recursive aggregation procedure starting from node i selects ℓ at round t is exactly

$$P_t(\ell \mid i) = \frac{w_t(\ell)}{\sum_{\ell' \in \mathcal{L}(i)} w_t(\ell')} = \frac{\exp(\eta_i^{\mathrm{shr}}\theta_{\ell,t})}{\sum_{\ell' \in \mathcal{L}(i)} \exp(\eta_i^{\mathrm{shr}}\theta_{\ell',t})}. \tag{7}$$

Consequently, viewing the safe subtree as an N_i-armed Exp3 problem over at most T rounds, we obtain

$$Y_\eta[i,T] \;\le\; Y_*[i,T] \;+\; 2\sqrt{T\,N_i \ln N_i}. \tag{8}$$

Proof. Since node i is safe, every non-leaf node in subtree(i) has gate $g(u) = \text{share}$, and every leaf in subtree(i) is shared. Hence the recursive aggregation rule is valid throughout the whole subtree rooted at i. The recursive aggregation algorithm selects a child j of an internal node u with probability

$$P_t(j \mid u) = \frac{W_t[j]}{W_t[u]}.$$

Fix any leaf $\ell \in \mathcal{L}(i)$, and let

$$i = u_0 \to u_1 \to \cdots \to u_m = \ell$$

be the unique path from node i to leaf ℓ. The probability that the recursive procedure starting at i finally reaches ℓ is the product of the transition probabilities along this path:

$$P_t(\ell \mid i) = \prod_{r=0}^{m-1} P_t(u_{r+1} \mid u_r) = \prod_{r=0}^{m-1} \frac{W_t[u_{r+1}]}{W_t[u_r]}.$$

This product telescopes:

$$P_t(\ell \mid i) = \frac{W_t[\ell]}{W_t[i]}.$$

Since a leaf has no descendants except itself,

$$W_t[\ell] = w_t(\ell) = \exp(\eta_i^{\mathrm{shr}}\theta_{\ell,t}),$$

and since i is the root of the current safe subtree,

$$W_t[i] = \sum_{\ell' \in \mathcal{L}(i)} w_t(\ell') = \sum_{\ell' \in \mathcal{L}(i)} \exp(\eta_i^{\mathrm{shr}}\theta_{\ell',t}).$$

Therefore,

$$P_t(\ell \mid i) = \frac{\exp(\eta_i^{\mathrm{shr}}\theta_{\ell,t})}{\sum_{\ell' \in \mathcal{L}(i)} \exp(\eta_i^{\mathrm{shr}}\theta_{\ell',t})},$$

which is exactly the arm-selection probability of an exponential-weights/Exp3 forecaster run directly on the leaf set $\mathcal{L}(i)$.

Hence, the safe subtree rooted at i can be viewed as an N_i-armed adversarial bandit problem. Since node i can be visited in at most T rounds, a valid upper bound is obtained by applying the standard Exp3 regret bound over horizon T with learning rate

$$\eta_i^{\mathrm{shr}} = \sqrt{\frac{\ln N_i}{T N_i}}.$$

This gives

$$Y_\eta[i,T] - Y_*[i,T] \;\le\; 2\sqrt{T\,N_i \ln N_i},$$

which proves the claim. □

Lemma RR-2 (R-Adjusted Local Risky Bounds). For any risky node i, the single-layer regret satisfies the following fixed-parameter bounds:

$$Y_\eta[i,T] \le \min_{j\in C_i} Y_\eta[j,T] + \big(d_i + \ln d_i\big)\, T^{\frac{R}{R+1}}, \qquad \text{if } C_i \subseteq \mathcal{L}, \tag{9}$$

$$Y_\eta[i,T] \le \min_{j\in C_i} Y_\eta[j,T] + \big(2d_i + \ln d_i\big)\, T^{\frac{R}{R+1}}, \qquad \text{if } C_i \not\subseteq \mathcal{L}. \tag{10}$$

Proof. Under the barrier algorithm, every barrier-free fully shared subtree is analyzed separately by Lemma RR-1, while every risky node follows the local ϵ-EXP3 dynamics. Hence, for a risky node i, the local regret is given by the Hou-style single-layer bound with local branching factor d_i:

$$\epsilon_i T + \frac{\ln d_i}{\eta_i^{\text{risk}}} + \eta_i^{\text{risk}} \sum_{t=1}^{T} \frac{d_i}{v[i,t]}.$$

Substituting the R-adjusted risky parameters

$$\eta_i^{\text{risk}} = T^{-\frac{R}{R+1}}, \qquad \epsilon_i = \begin{cases} 0, & C_i \subseteq \mathcal{L}, \\ d_i\, T^{-\frac{1}{R+1}}, & C_i \not\subseteq \mathcal{L}, \end{cases}$$

and using the same visit-probability scaling argument as in the original ϵ-EXP3 analysis, but with the risky skeleton depth R in place of the full depth L, yields the stated bounds. When $C_i \subseteq \mathcal{L}$, the exploration term vanishes, so the local contribution is $(d_i + \ln d_i)T^{\frac{R}{R+1}}$. Otherwise, the extra education/exploration contribution adds another $d_i T^{\frac{R}{R+1}}$, giving $(2d_i + \ln d_i)T^{\frac{R}{R+1}}$. □

Theorem RR-1 (Recursive Root Regret Bound of Risky-PS). Define recursively, for every node i,

$$\Delta(i) \;\triangleq\; \begin{cases} 0, & i \in \mathcal{L}, \\ 2\sqrt{T\, N_i \ln N_i}, & i \text{ is safe}, \\ \big(d_i + \ln d_i\big)\, T^{\frac{R}{R+1}}, & i \text{ is risky and } C_i \subseteq \mathcal{L}, \\ \big(2d_i + \ln d_i\big)\, T^{\frac{R}{R+1}} + \min\limits_{j\in C_i^*} \Delta(j), & i \text{ is risky and } C_i \not\subseteq \mathcal{L}. \end{cases} \tag{11}$$

Then for every node i,

$$Y_\eta[i,T] \;\le\; Y_*[i,T] + \Delta(i). \tag{12}$$

In particular, for the root node r,

$$\text{Reg}_{\text{root}}(T) \;\triangleq\; Y_\eta[r,T] - Y_*[r,T] \;\le\; \Delta(r). \tag{13}$$

Proof. We proceed by structural induction on the subtree rooted at i.

Base case: If $i \in \mathcal{L}$, then by definition

$$Y_\eta[i,T] = Y_*[i,T],$$

and since $\Delta(i) = 0$, (12) holds.

Safe case: If i is safe, then (12) follows directly from Lemma RR-1:

$$Y_\eta[i,T] \le Y_*[i,T] + 2\sqrt{T\, N_i \ln N_i} = Y_*[i,T] + \Delta(i).$$

Risky leaf-parent case: If i is risky and $C_i \subseteq \mathcal{L}$, then Lemma RR-2 gives

$$Y_\eta[i,T] \le \min_{j\in C_i} Y_\eta[j,T] + \big(d_i + \ln d_i\big)\, T^{\frac{R}{R+1}}.$$

Since all children are leaves,

$$\min_{j\in C_i} Y_\eta[j,T] = \min_{j\in C_i} Y_*[j,T] = Y_*[i,T].$$

Therefore,

$$Y_\eta[i,T] \le Y_*[i,T] + \big(d_i + \ln d_i\big)\, T^{\frac{R}{R+1}} = Y_*[i,T] + \Delta(i).$$

Risky non-leaf case: Suppose now that i is risky and $C_i \not\subseteq \mathcal{L}$. By Lemma RR-2,

$$Y_\eta[i,T] \le \min_{j\in C_i} Y_\eta[j,T] + \big(2d_i + \ln d_i\big)\, T^{\frac{R}{R+1}}. \tag{14}$$

Take any optimal child $j^* \in C_i^*$. By definition of C_i^*,

$$Y_*[j^*,T] = Y_*[i,T]. \tag{15}$$

By the induction hypothesis applied to node j^*,

$$Y_\eta[j^*,T] \le Y_*[j^*,T] + \Delta(j^*). \tag{16}$$

Also,

$$\min_{j\in C_i} Y_\eta[j,T] \le Y_\eta[j^*,T]. \tag{17}$$

Substituting (16) and (17) into (14), and then using (15), yields

$$Y_\eta[i,T] \le Y_*[i,T] + \Delta(j^*) + \big(2d_i + \ln d_i\big)\, T^{\frac{R}{R+1}}.$$

Since this holds for any $j^* \in C_i^*$, taking the minimum over C_i^* gives

$$Y_\eta[i,T] \le Y_*[i,T] + \min_{j\in C_i^*} \Delta(j) + \big(2d_i + \ln d_i\big)\, T^{\frac{R}{R+1}} = Y_*[i,T] + \Delta(i).$$

Thus the induction closes, and (12) holds for all nodes i. Applying it to the root proves (13). □

Sanity checks.

- **All-share.** If every non-leaf node has gate $g(i) =$ share and all leaves are shared, then the whole tree is safe, so r is safe and
$$\Delta(r) = 2\sqrt{T\,|\mathcal{L}|\,\ln|\mathcal{L}|},$$
which is independent of the depth L.

- **All-unshare.** If every internal node has gate $g(i) =$ unshare, then every internal node is risky and there is no barrier-free shared suffix, so $R = L$. Hence the recursive expansion yields
$$\Delta(r) = \left(\sum_{i\in\mathcal{P}_{\text{int}}(r)} d_i + \sum_{i\in\mathcal{P}_{\text{int}}(r)\setminus\{i_{\text{leaf-parent}}\}} d_i + \sum_{i\in\mathcal{P}_{\text{int}}(r)} \ln d_i\right) T^{\frac{L}{L+1}},$$
where $\mathcal{P}_{\text{int}}(r)$ denotes the set of internal nodes on the optimal root-to-leaf path, and $i_{\text{leaf-parent}}$ is its last internal node. In the homogeneous case $d_i \equiv D$, this simplifies to
$$\Delta(r) = \big((2L-1)D + L\ln D\big)\, T^{\frac{L}{L+1}},$$
which exactly recovers the original worst-case endpoint.

2 Recursive Root Regret Bound for Risky-PS

Definitions

For any node i, define the cumulative expected loss of the algorithm starting from i by

$$Y_\eta[i,T] \;\triangleq\; \sum_{t=1}^{T} \mathbb{E}[\, y_\eta[i,t]\,].$$

Let $Y_*[i,T]$ denote the cumulative loss of the optimal stationary policy starting from node i, defined recursively as

$$Y_*[i,T] \;=\; \begin{cases} \sum_{t=1}^{T} c_t(i), & \text{if } i \in \mathcal{L}, \\ \min_{j\in C_i} Y_*[j,T], & \text{otherwise.} \end{cases}$$

For any non-leaf node i, define its local branching factor and optimal-child set by

$$d_i \;\triangleq\; |C_i|, \qquad C_i^* \;\triangleq\; \arg\min_{j\in C_i} Y_*[j,T].$$

To make the risky depth precise, define the *risky height* of each node i recursively by

$$R(i) \;\triangleq\; \begin{cases} 0, & \text{if } i \text{ is safe,} \\ 1, & \text{if } i \text{ is risky and } C_i \subseteq \mathcal{L}, \\ 1 + \max_{j\in C_i} R(j), & \text{if } i \text{ is risky and } C_i \not\subseteq \mathcal{L}. \end{cases}$$

We then define the *risky skeleton depth* of the whole tree as

$$R \;\triangleq\; R(r),$$

where r is the root. Equivalently, R is the maximum number of risky nodes that can appear on any root-to-leaf path before the path enters a fully safe suffix.

For risky nodes, we use the R-adjusted parameters

$$\eta_i^{\text{risk}} \;=\; T^{-\frac{R}{R+1}}, \qquad \epsilon_i \;=\; \begin{cases} 0, & C_i \subseteq \mathcal{L}, \\ d_i\, T^{-\frac{1}{R+1}}, & C_i \not\subseteq \mathcal{L}. \end{cases}$$

For a safe node i, let

$$N_i \;\triangleq\; |\mathcal{L}(i)|$$

be the number of leaves in the safe subtree rooted at i. We upper bound its local contribution by the standard N_i-armed Exp3 rate over at most T rounds, using

$$\eta_i^{\text{shr}} \;=\; \sqrt{\frac{\ln N_i}{T\, N_i}}.$$

Lemma RR-1 (Safe-Subtree Exp3 Equivalence). For any safe node i, let

$$N_i \;\triangleq\; |\mathcal{L}(i)|$$

denote the number of leaves in the safe subtree rooted at i. Then the recursive selection rule of the full-share algorithm restricted to the subtree of i is equivalent to running an N_i-armed Exp3 forecaster directly on the leaf set $\mathcal{L}(i)$.

More precisely, if the leaf weights are

$$w_t(\ell) \;\triangleq\; \exp\!\big(\eta_i^{\text{shr}}\theta_{\ell,t}\big), \qquad \ell \in \mathcal{L}(i),$$

and the recursive aggregate at any node u is

$$W_t[u] \;\triangleq\; \sum_{\ell\in\text{subtree}(u)} w_t(\ell),$$

then for any leaf $\ell \in \mathcal{L}(i)$, the probability that the recursive aggregation procedure starting from node i selects ℓ at round t is exactly

$$P_t(\ell \mid i) = \frac{w_t(\ell)}{\sum_{\ell' \in \mathcal{L}(i)} w_t(\ell')} = \frac{\exp(\eta_i^{\text{shr}} \theta_{\ell,t})}{\sum_{\ell' \in \mathcal{L}(i)} \exp(\eta_i^{\text{shr}} \theta_{\ell',t})}. \tag{7}$$

Consequently, viewing the safe subtree as an N_i-armed Exp3 problem over at most T rounds, we obtain

$$Y_\eta[i,T] \;\le\; Y_*[i,T] \;+\; 2\sqrt{T\, N_i \ln N_i}. \tag{8}$$

Proof. Since node i is safe, every leaf in subtree(i) is shared. The recursive aggregation algorithm selects a child j of an internal node u with probability

$$P_t(j \mid u) = \frac{W_t[j]}{W_t[u]}.$$

Fix any leaf $\ell \in \mathcal{L}(i)$, and let

$$i = u_0 \to u_1 \to \cdots \to u_m = \ell$$

be the unique path from node i to leaf ℓ. The probability that the recursive procedure starting at i finally reaches ℓ is the product of the transition probabilities along this path:

$$P_t(\ell \mid i) = \prod_{r=0}^{m-1} P_t(u_{r+1} \mid u_r) = \prod_{r=0}^{m-1} \frac{W_t[u_{r+1}]}{W_t[u_r]}.$$

This product telescopes:

$$P_t(\ell \mid i) = \frac{W_t[\ell]}{W_t[i]}.$$

Since a leaf has no descendants except itself,

$$W_t[\ell] = w_t(\ell) = \exp(\eta_i^{\text{shr}} \theta_{\ell,t}),$$

and since i is the root of the current safe subtree,

$$W_t[i] = \sum_{\ell' \in \mathcal{L}(i)} w_t(\ell') = \sum_{\ell' \in \mathcal{L}(i)} \exp(\eta_i^{\text{shr}} \theta_{\ell',t}).$$

Therefore,

$$P_t(\ell \mid i) = \frac{\exp(\eta_i^{\text{shr}} \theta_{\ell,t})}{\sum_{\ell' \in \mathcal{L}(i)} \exp(\eta_i^{\text{shr}} \theta_{\ell',t})},$$

which is exactly the arm-selection probability of an exponential-weights/Exp3 forecaster run directly on the leaf set $\mathcal{L}(i)$.

Hence, the safe subtree rooted at i can be viewed as an N_i-armed adversarial bandit problem. Since node i can be visited in at most T rounds, a valid upper bound is obtained by applying the standard Exp3 regret bound over horizon T with learning rate

$$\eta_i^{\text{shr}} = \sqrt{\frac{\ln N_i}{T N_i}}.$$

This gives

$$Y_\eta[i,T] - Y_*[i,T] \;\le\; 2\sqrt{T\, N_i \ln N_i},$$

which proves the claim. □

Lemma RR-2 (R-Adjusted Local Risky Bounds). For any risky node i, the single-layer regret satisfies the following fixed-parameter bounds:

$$Y_\eta[i,T] \le \min_{j \in C_i} Y_\eta[j,T] + \big(d_i + \ln d_i\big)\, T^{\frac{R}{R+1}}, \qquad \text{if } C_i \subseteq \mathcal{L}, \tag{9}$$

$$Y_\eta[i,T] \le \min_{j \in C_i} Y_\eta[j,T] + \big(2d_i + \ln d_i\big)\, T^{\frac{R}{R+1}}, \qquad \text{if } C_i \not\subseteq \mathcal{L}. \tag{10}$$

Proof. Under the truncated algorithm, safe subtrees are analyzed separately by Lemma RR-1, while every risky node follows the local ϵ-EXP3 dynamics. Hence, for a risky node i, the local regret is given by the Hou-style single-layer bound with local branching factor d_i:

$$\epsilon_i T + \frac{\ln d_i}{\eta_i^{\text{risk}}} + \eta_i^{\text{risk}} \sum_{t=1}^{T} \frac{d_i}{v[i,t]}.$$

Substituting the R-adjusted risky parameters

$$\eta_i^{\text{risk}} = T^{-\frac{R}{R+1}}, \qquad \epsilon_i = \begin{cases} 0, & C_i \subseteq \mathcal{L}, \\ d_i\, T^{-\frac{1}{R+1}}, & C_i \not\subseteq \mathcal{L}, \end{cases}$$

and using the same visit-probability scaling argument as in the original ϵ-EXP3 analysis, but with the risky skeleton depth R in place of the full depth L, yields the stated bounds. When $C_i \subseteq \mathcal{L}$, the exploration term vanishes, so the local contribution is $(d_i + \ln d_i)T^{\frac{R}{R+1}}$. Otherwise, the extra education/exploration contribution adds another $d_i T^{\frac{R}{R+1}}$, giving $(2d_i + \ln d_i)T^{\frac{R}{R+1}}$. □

Theorem RR-1 (Recursive Root Regret Bound of Risky-PS). Define recursively, for every node i,

$$\Delta(i) \;\triangleq\; \begin{cases} 0, & i \in \mathcal{L}, \\ 2\sqrt{T\, N_i \ln N_i}, & i \text{ is safe}, \\ \big(d_i + \ln d_i\big)\, T^{\frac{R}{R+1}}, & i \text{ is risky and } C_i \subseteq \mathcal{L}, \\ \big(2d_i + \ln d_i\big)\, T^{\frac{R}{R+1}} + \min\limits_{j \in C_i^*} \Delta(j), & i \text{ is risky and } C_i \not\subseteq \mathcal{L}. \end{cases} \tag{11}$$

Then for every node i,

$$Y_\eta[i,T] \;\le\; Y_*[i,T] + \Delta(i). \tag{12}$$

In particular, for the root node r,

$$\text{Reg}_{\text{root}}(T) \;\triangleq\; Y_\eta[r,T] - Y_*[r,T] \;\le\; \Delta(r). \tag{13}$$

Proof. We proceed by structural induction on the subtree rooted at i.

Base case: If $i \in \mathcal{L}$, then by definition

$$Y_\eta[i,T] = Y_*[i,T],$$

and since $\Delta(i) = 0$, (12) holds.

Safe case: If i is safe, then (12) follows directly from Lemma RR-1:

$$Y_\eta[i,T] \le Y_*[i,T] + 2\sqrt{T\, N_i \ln N_i} = Y_*[i,T] + \Delta(i).$$

Risky leaf-parent case: If i is risky and $C_i \subseteq \mathcal{L}$, then Lemma RR-2 gives

$$Y_\eta[i,T] \le \min_{j \in C_i} Y_\eta[j,T] + \big(d_i + \ln d_i\big)\, T^{\frac{R}{R+1}}.$$

Since all children are leaves,

$$\min_{j \in C_i} Y_\eta[j,T] = \min_{j \in C_i} Y_*[j,T] = Y_*[i,T].$$

Therefore,

$$Y_\eta[i,T] \le Y_*[i,T] + \big(d_i + \ln d_i\big)\, T^{\frac{R}{R+1}} = Y_*[i,T] + \Delta(i).$$

Risky non-leaf case: Suppose now that i is risky and $C_i \not\subseteq \mathcal{L}$. By Lemma RR-2,

$$Y_\eta[i,T] \le \min_{j \in C_i} Y_\eta[j,T] + \big(2d_i + \ln d_i\big)\, T^{\frac{R}{R+1}}. \tag{14}$$

Take any optimal child $j^* \in C_i^*$. By definition of C_i^*,

$$Y_*[j^*,T] = Y_*[i,T]. \tag{15}$$

By the induction hypothesis applied to node j^*,

$$Y_\eta[j^*, T] \le Y_*[j^*, T] + \Delta(j^*). \tag{16}$$

Also,

$$\min_{j \in C_i} Y_\eta[j, T] \le Y_\eta[j^*, T]. \tag{17}$$

Substituting (16) and (17) into (14), and then using (15), yields

$$Y_\eta[i, T] \le Y_*[i, T] + \Delta(j^*) + \big(2d_i + \ln d_i\big)\, T^{\frac{R}{R+1}}.$$

Since this holds for any $j^* \in C_i^*$, taking the minimum over C_i^* gives

$$Y_\eta[i, T] \le Y_*[i, T] + \min_{j \in C_i^*} \Delta(j) + \big(2d_i + \ln d_i\big)\, T^{\frac{R}{R+1}} = Y_*[i, T] + \Delta(i).$$

Thus the induction closes, and (12) holds for all nodes i. Applying it to the root proves (13). □

Sanity checks.

- **All-share.** If the whole tree is safe, then r is safe and

  $$\Delta(r) = 2\sqrt{T\,|\mathcal{L}|\,\ln|\mathcal{L}|},$$

  which is independent of the depth L.

- **All-unshare.** If all leaves are unshared, then every internal node is risky and there is no safe suffix, so $R = L$. Hence the recursive expansion yields

  $$\Delta(r) = \left(\sum_{i \in \mathcal{P}_{\text{int}}(r)} d_i + \sum_{i \in \mathcal{P}_{\text{int}}(r) \setminus \{i_{\text{leaf-parent}}\}} d_i + \sum_{i \in \mathcal{P}_{\text{int}}(r)} \ln d_i \right) T^{\frac{L}{L+1}},$$

  where $\mathcal{P}_{\text{int}}(r)$ denotes the set of internal nodes on the optimal root-to-leaf path, and $i_{\text{leaf-parent}}$ is its last internal node. In the homogeneous case $d_i \equiv D$, this simplifies to

  $$\Delta(r) = \big((2L-1)D + L\ln D\big)\, T^{\frac{L}{L+1}},$$

  which exactly recovers the original worst-case endpoint.

3 All-Share Lower Bound via Adversarial Bandit Reduction

Assume the whole tree is all-share, so that the root r is safe. Let

$$N \triangleq |\mathcal{L}(r)|$$

be the number of leaves. Then the recursive full-share selection rule induces an N-armed adversarial bandit policy on the leaf set $\mathcal{L}(r)$. Consequently, there exists a universal constant $c_0 > 0$ such that for any $\delta \in (0, 1)$ satisfying the conditions of Theorem 17.4, there exists a cost sequence $\{c_t(\ell)\}_{t \le T, \ell \in \mathcal{L}(r)} \subset [0, 1]$ such that

$$\mathbb{P}\left(\mathrm{Reg}_{\text{root}}(T) \;\ge\; c_0 \sqrt{T\,N\,\log \frac{1}{2\delta}} \right) \;\ge\; \delta. \tag{LB-1}$$

In particular, there exists a universal constant $c_1 > 0$ such that

$$\sup_{c_{1:T}} \mathbb{E}[\mathrm{Reg}_{\text{root}}(T)] \;\ge\; c_1 \sqrt{T\,N}. \tag{LB-2}$$

Proof. For any leaf $\ell \in \mathcal{L}(r)$, let

$$r = u_0 \to u_1 \to \cdots \to u_m = \ell$$

be the unique root-to-leaf path. Under the recursive aggregation rule,

$$P_t(u_{s+1} \mid u_s) = \frac{W_t[u_{s+1}]}{W_t[u_s]},$$

hence the probability of selecting leaf ℓ at round t is

$$P_t(\ell \mid r) = \prod_{s=0}^{m-1} \frac{W_t[u_{s+1}]}{W_t[u_s]} = \frac{W_t[\ell]}{W_t[r]} = \frac{w_t(\ell)}{\sum_{\ell' \in \mathcal{L}(r)} w_t(\ell')}.$$

Therefore, the recursive full-share algorithm on the all-share tree is equivalent, at the leaf level, to an N-armed bandit policy over $\mathcal{L}(r)$.

Now define rewards by

$$x_{t\ell} \triangleq 1 - c_t(\ell), \qquad \ell \in \mathcal{L}(r).$$

Then the reward regret and the cost regret coincide:

$$\max_{\ell \in \mathcal{L}(r)} \sum_{t=1}^{T} (x_{t\ell} - x_{tA_t}) = \sum_{t=1}^{T} c_t(A_t) - \min_{\ell \in \mathcal{L}(r)} \sum_{t=1}^{T} c_t(\ell) = \mathrm{Reg}_{\mathrm{root}}(T).$$

Applying Theorem 17.4 with $k = N$ and $n = T$ yields the high-probability lower bound (LB-1). Choosing a fixed constant δ (e.g. $\delta = 1/4$) gives (LB-2). $\square$